

⇐ حل امتحان الفيزياء [تقنيات عددية]

⇒ Q1:- المسألة

□ ⇒ $h = 0.3$, $\frac{dy}{dx} + 5y^2 = \sin x$, $y(0.3) = 5$, $y(0.6) = ??$

⇒ sol:- $\frac{dy}{dx} = \sin x - 5y^2$, (حلول الأوتار لا (Red))

⇒ $y_{i+1} = y_i + f(x_i, y_i)h \rightarrow$ (من الشيفرة)

↓
⇒ $y(0.6) = \text{5} + f(0.3, 5) * 0.3$

$= 5 + [-124.7 * 0.3]$

$= -32.41 \rightarrow \textcircled{A}$

⇒ □ ⇒ $\sqrt{x} = x \Rightarrow x^2 = 7 \Rightarrow f(x) = x^2 - 7$, $x_i = 2$ [من الشيفرة]

⇒ $x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)} \rightarrow$ (من الشيفرة)

⇒ $f(2) = 4 - 7 = -3$ / $f'(x) = 2x \Rightarrow f'(2) = 4$

⇒ $x_i = 2 - \frac{(-3)}{4} = 2.75 \rightarrow \textcircled{C}$

##

③ $\Rightarrow \epsilon_a = 0.004\% \Rightarrow \epsilon_s = (0.5 \times 10^{2-n})\% \rightarrow$ (الشك) 6

$\Rightarrow 0.004\% = (0.5 \times 10^{2-n})\% \rightarrow$ (عدد الألف)

$\Rightarrow \boxed{n = 4} \rightarrow$ ③

④ $\Rightarrow$ solve $x + 2y = 12 / 2x + ky = 22, k = ? \rightarrow$ no. sol...

$\Rightarrow$ no. solution $\xrightarrow{\text{مساواة}} \begin{bmatrix} - & - \\ 0 & 0 \\ \text{zero} & \text{zero} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} - \\ - \end{bmatrix}$

$\Rightarrow \begin{bmatrix} 1 & 2 \\ 2 & k \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 12 \\ 22 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 2 \\ 2 & k \end{bmatrix} \xrightarrow{-2R_1 + R_2} \begin{bmatrix} 1 & 2 \\ 0 & k-4 \end{bmatrix}$

$\Rightarrow \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 12 \\ 2 \end{bmatrix} \Rightarrow k = 4 \rightarrow$ ⑤

⑤ $\Rightarrow$ solve $\sum_{i=1}^4 \sum_{j=1}^4 \left(\frac{x}{y^2} \right) \cdot \delta x \delta y \Rightarrow h=1, k=1$

$\Rightarrow$

x/y	2	3	4
2	$\boxed{0.5}$	$\bigcirc \frac{3}{4}$	$\boxed{1}$
3	$\bigcirc \frac{2}{9}$	$\triangle \frac{1}{3}$	$\bigcirc \frac{4}{9}$
4	$\boxed{\frac{1}{8}}$	$\bigcirc \frac{3}{16}$	$\boxed{\frac{1}{4}}$

$\left(\frac{x}{y^2} \right) \rightarrow$ مساحة العناصر
وضوح القايير
من (2-4)

$h=2, k=2 \leftarrow$ ريمان

x/y	2	4
2	$\boxed{0.5}$	$\boxed{1}$
4	$\boxed{\frac{1}{8}}$	$\boxed{\frac{1}{4}}$

$\Rightarrow \frac{2 \times 2}{4} [1.875 + 0 + 0] = 1.875 \rightarrow$
 $\leftarrow$ كالتقريب

$\Rightarrow = \frac{hk}{4} [\text{sum } \square + 2(\text{sum } \bigcirc) + 4(\text{sum } \triangle)]$

$= \frac{1}{4} \left[\frac{15}{8} + [2 * \frac{77}{48}] + [4 * \frac{1}{3}] \right] = \boxed{1.604}$

الجواب منه موجود بالتقريبات (السؤال غير متطابق بالتقريبات من الجوابين هذا خطأ لنا)

6 $\Rightarrow g(x) = \cos(x) - 0.5 e^x$ / لازم رقمی می باشد
و لا یسر (-) سالب

$\Rightarrow [0, 0.7]$ $\Rightarrow g(0) = \cos(0) - 0.5 e^0 \Rightarrow g(0) = 0.5 \rightarrow (+)$
 $\Rightarrow g(0.7) = \cos(0.7) - 0.5 e^{0.7} \Rightarrow g(0.7) = -0.242 \rightarrow (-)$
 $\downarrow$
 $\checkmark$ ①

$\Rightarrow$ ② $\Rightarrow$ solve $[2, 4]$, $\epsilon_e = 0.00583$

$\Rightarrow n \geq \frac{\ln(b-a) - \ln(\epsilon_e)}{\ln(2)} \rightarrow$ (مدت التكرار)

$\Rightarrow n \geq \frac{\ln(4-2) - \ln(0.00583)}{\ln(2)} \Rightarrow n \geq 8.4223$

$\Rightarrow n \geq 9 \rightarrow$ ①

ما را در حالت
تکرار من (8)
منقذ بها لأقدم عدد
الحاصل

$\Rightarrow$ ③ $\Rightarrow$ solve ④ $\rightarrow$ point (نقطه) $\rightarrow$ center . d . A

$f(x) = \frac{2^x}{x}$, $x_i = 3$, $h = 1$.

$\Leftarrow$ به ازای نقاط و قانون (center) $\Leftarrow$ مستخدم الا نه من التالي : (من الشيفرة)

$f'(x_i) = \frac{-f(x_{i+2}) + 8f(x_{i+1}) - 8f(x_{i-1}) + f(x_{i-2}))}{12h}$

اینجا اشتباهه
لأنه بدو
نقاط

$\Rightarrow x_i = 3, x_{i+1} = 4, x_{i+2} = 5, x_{i-1} = 2, x_{i-2} = 1 \rightarrow$

$\Rightarrow f'(x_i) = \frac{-f(5) + 8f(4) - 8f(2) + f(1)}{12} = 0.967 \rightarrow$ ②

اینجا باید دو نقطه
بگذار (h)

$$\textcircled{9} \Rightarrow \text{sol'n} \int_{8.1}^{8.5} \sqrt{x \sin(x)} \cdot dx' \Rightarrow h = 0.2$$

$$[x_0 = 8.1, x_1 = 8.3, x_2 = 8.5]$$

$$\Rightarrow h = \frac{8.5 - 8.1}{0.2} = 2 \rightarrow \text{Simpson Rule (simple)}$$

$$\Rightarrow \int_{x_0}^{x_2} f(x) \cdot dx \approx \frac{h}{3} [f(x_0) + 4f(x_1) + f(x_2)]$$

$$\approx \frac{0.2}{3} [f(8.1) + [4 * f(8.3)] + f(8.5)]$$

$$\approx \frac{0.2}{3} [7.8561 + [4 * 7.433] + 6.78714]$$

$$\approx 2.973 \rightarrow \textcircled{A}$$

$$\boxed{10} \Rightarrow \text{sol'n} f(x) = 2^x, \text{ points} \rightarrow 0, 1, 2$$

نقاط 3 نقاط $\therefore$ الإقتراح لدرجة 2 (من الدرجة 2)
 عدد النقاط - 1 = درجة الإقتراح

$$\Rightarrow f(0.333) = \frac{(0.333-1)(0.333-2)}{(0-1)(0-2)} f(0) + \frac{(0.333-0)(0.333-2)}{(1-0)(1-2)} f(1)$$

$$+ \frac{(0.333-0)(0.333-1)}{(2-0)(2-1)} f(2) = 1.222 \rightarrow \textcircled{A}$$

$$\boxed{11} \Rightarrow [3A]^{-1} \xrightarrow{\text{inverse}} \frac{1}{3} A^{-1}$$

$$\boxed{12} \Rightarrow \text{sol'n} f(x, y) = y^2 x^2, (1, 2)$$

$$\Rightarrow \frac{\partial f}{\partial x} = 2xy^2, \frac{\partial f}{\partial y} = 2yx^2$$

$$\Rightarrow (1, 2) \Rightarrow \nabla f = \begin{bmatrix} 8 \\ 4 \end{bmatrix} \Rightarrow = 8\hat{i} + 4\hat{j} \quad \textcircled{B}$$

⇒ 13 ⇒ solve $x + 4y + 9z = 20$

⇒ $z = \frac{20 - x - 4y}{9}$, $x = 3.63$, $y = 1.718$

⇒ $z = \frac{20 - 3.63 - (4 * 1.718)}{9} = 1.055 \rightarrow \textcircled{A}$

⇒ 14 ⇒ solve $\frac{dy}{dx} = x + y^2$, $y(0.2) = 2.34$

⇒ $h = 0.1$ / ⇒ $k_1 \rightarrow$ Runge-Kutta $\rightarrow 4^{th}$ order

⇒ $k_1 = h * f(x_i, y_i) \rightarrow$ (مشتق)

مشتق
من القيمة
(1.)

⇒ $k_1 = 0.1 * f(0.2, 2.34)$

$= 0.1 * 5.67 = 0.567 \approx 0.566 \rightarrow \textcircled{B}$

⇒ 15 ⇒ solve ⇒ minimum / $f(x) = \sin(x) + x^2$ / $[x_L, x_u] = [-0.8, 0]$

⇒ $d = 0.618033 (x_u - x_L) = 0.618033 * (0 - (-0.8)) = 0.4944264$

⇒ $x_1 = x_L + d = -0.8 + 0.4944264 = -0.3055736$

⇒ $x_2 = x_u - d = 0 - 0.4944264 = -0.4944264$

⇒ $f(x_1) = -0.207465$ / $f(x_2) = -0.230069$

⇒ minimum ⇒ $f(x_1) > f(x_2) \Rightarrow x_1 = x_u$

⇒ $[x_L = -0.8 / x_u = x_1 = -0.3056] \Rightarrow \textcircled{B}$

⇒ Q.2 ⇒ solve extreme value ~~max/min~~

⇒

x	f(x)	b ₁	b ₂
2	5	-10	3
3	-5	-1	
5	-7		

$$\Rightarrow b_1 = f(3, 2) = \frac{f(3) - f(2)}{3 - 2} = \frac{-5 - 5}{1} = -10$$

$$\Rightarrow b'_1 = f(5, 3) = \frac{f(5) - f(3)}{5 - 3} = \frac{-7 + 5}{2} = \frac{-2}{2} = -1$$

$$\Rightarrow b_2 = f(5, 3, 2) = \frac{-1 + 10}{5 - 2} = \frac{9}{3} = 3$$

$$\Rightarrow f(x) = b_0 + b_1(x - x_0) + b_2(x - x_0)(x - x_1)$$

$$\Rightarrow f(x) = 5 + 10(x - 2) + 3(x - 2)(x - 3)$$

$$= 5 - 10x + 20 + 3(x^2 - 3x - 2x + 6)$$

$$= 5 - 10x + 20 + 3x^2 - 15x + 18$$

$$f(x) = 3x^2 - 25x + 43$$

$$f'(x) = 6x - 25 \Rightarrow 6x - 25 = 0 \Rightarrow x = 4.167 \quad \#$$

$$\Rightarrow f''(x) = 6 \Rightarrow f''(4.167) = 6 \quad \#$$

2.3.1

$$\Rightarrow \int_2^8 \frac{9 \sin(x)}{x} \cdot dx. \quad / \Rightarrow n=1 \quad / \Rightarrow \text{second order}$$

clear data $\leftarrow (4, 2)$

Trapezoidal

$$\Rightarrow n=1 \Rightarrow h = \frac{b-a}{n} = \frac{8-2}{1} = 6 \Rightarrow \hat{I} = \frac{6}{2} [f(2) + f(8)]$$

$$\Rightarrow R(1,0) = 3 [4.09184 + 1.11303] = 15.61461$$

$$\Rightarrow n=2 \Rightarrow h = \frac{8-2}{2} = 3 \Rightarrow R(2,0) = \frac{3}{2} [f(2) + 2f(5) + f(8)]$$

$$\Rightarrow R(2,0) = \frac{3}{2} [4.09184 + [2 \times -1.72606] + 1.11303] = 2.62913$$

$$\Rightarrow n=4 \Rightarrow h = \frac{8-2}{4} = \frac{6}{4} \Rightarrow R(3,0) = \frac{6}{2} [f(2) + 2f(3.5) + f(5) + 4f(6)]$$

$$+ f(8)] = \frac{6}{2} [4.09184 + [2 \times (-0.90201) + -1.72606 + 0.29786]$$

$$+ 1.11303] = \text{~~0.40834~~} 0.40834$$

$$\Rightarrow n=6 \Rightarrow h = \frac{8-2}{6} = 1 \Rightarrow R(4,0) = \frac{1}{2} [f(2) + [2f(3) + f(4) + f(5) + f(6) + f(7)]$$

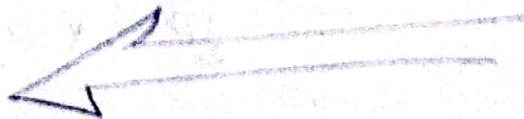
$$+ f(8)]$$

$$\Rightarrow R(4,0) = \frac{1}{2} [4.09184 + [2(0.42336 + -1.70281 + -1.72606 + 0.41912$$

$$+ 0.84470) + 1.11303] = 0.045755 \quad \times$$

{ $\frac{1}{4^k}$ rule}

$$[R(i,k) = \frac{4^k R(i,k-1) - R(i,k-1)}{4^k - 1}]$$



$R(i, k)$	First order	Second order
$[15.614613]$ $\hat{\uparrow}$ $R(1,0)$		
$[2.62913]$ $\hat{\uparrow}$ $R(2,0)$	-1.699363333	
$[0.40834]$ $\hat{\uparrow}$ $R(3,0)$	-0.3319233333	-0.2407543333
$[0.045755]$ $\hat{\uparrow}$ $R(4,0)$	-0.07510666667	-0.05798933333

$$\Rightarrow R(2,1) = \frac{4^{(1)} * R(2,0) - R(1,0)}{4^{(1)} - 1} = -1.69936$$

$$\Rightarrow R(3,1) = \frac{4^{(1)} * R(3,0) - R(2,0)}{4^{(0)} - 1} = -0.33192$$

$$\Rightarrow R(4,1) = \frac{4^{(1)} * R(4,0) - R(3,0)}{4^{(1)} - 1} = -0.07511$$

$$\Rightarrow R(3,2) = \frac{4^{(2)} * R(3,1) - R(2,1)}{4^{(2)} - 1} = -0.24076$$

$$\Rightarrow R(4, 2) = \frac{4^{(2)} * R(4, 1) - R(3, 1)}{4^{(2)} - 1} = -0.05799$$

ابراهيم
XXXX

⇒ (Q. 2)

⇒ sol:-

$$f(x) = e^{-x} \cos(x) \quad / \quad x = \frac{\pi}{4} \quad / \quad h = \frac{\pi}{12}$$

$$\ddot{f}(x_i) = \frac{f(x_{i+1}) - 2f(x_i) + f(x_{i-1}))}{h^2}$$

$$\Rightarrow x_i = \frac{\pi}{4}, \quad x_{i+1} = \frac{3\pi}{4} + \frac{\pi}{12} = \frac{4\pi}{12} = \frac{\pi}{3} \quad / \quad x_{i-1} = \frac{\pi}{4} - \frac{\pi}{12} = \frac{2\pi}{12} = \frac{\pi}{6}$$

$$\Rightarrow \ddot{f}\left(\frac{\pi}{4}\right) = \frac{f\left(\frac{\pi}{3}\right) - 2f\left(\frac{\pi}{4}\right) + f\left(\frac{\pi}{6}\right)}{\left(\frac{\pi}{12}\right)^2}$$

$$\Rightarrow \ddot{f}\left(\frac{\pi}{4}\right) = \frac{(\cancel{0.17546}) - (2 \times 0.32239) + 0.51302}{\left(\frac{\pi}{12}\right)^2}$$

$$\Rightarrow \ddot{f}\left(\frac{\pi}{4}\right) = \boxed{0.63759}$$

✗

Ans

[Q.4]

⇒ sol

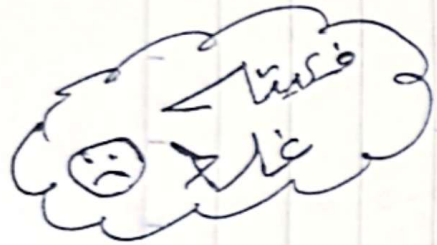
x	1	2	3	4	5	6
y	-2	1	10	29	62	115

$$n = 6 - 1 = 5 \rightarrow \underline{b_5}$$

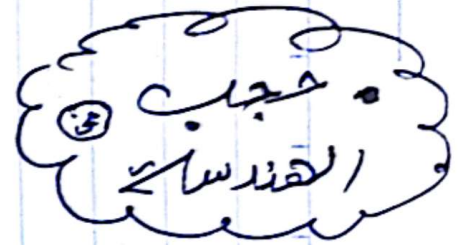
	x	y	b ₁	b ₂	b ₃	b ₄	b ₅
x ₀	1	-2	3	3	0.6667	0	0.01665
x ₁	2	1	9	5	0.6667	0.01665	0.01665
x ₂	3	10	19	7	0.08332	0.08332	
x ₃	4	29	33	10			
x ₄	5	62	53				
x ₅	6	115					

$$f_5(x) = -2 + 3(x-1) + 3(x-1)(x-2) + 0.667(x-1)(x-2)(x-3) + 0(x-1)(x-2)(x-3)(x-4) + 0.01665(x-1)(x-2)(x-3)(x-4)(x-5)$$

$$= -2 + 3x - 3 + 3(x^2 - 3x + 2) + 0.667(x^3 - 3x^2 + 2x - 3x^2 + 6x + 3x - 6) - 0.01665$$



$$\rightarrow (x^4 - 3x^3 + 2x^2 - 4x^3 + 12x^2 - 8x)(x-5)$$



$$= -2 + 3x - 3 + 3x^2 - 6x - 3x + 6 + 0.6667x^3 - 2.0001x^2 + 1.3334x - 2.0001x^2 + 4.0002x + 2.0001x - 4.0002$$

$$\rightarrow -0.01665(x^5 - 3x^4 + 2x^3 - 4x^4 + 12x^3 - 8x^2 + 5x^4 - 5x^3 + 10x^2 - 20x^3 + 60x^2 - 40x)$$

$$= -2 + 3x - 3 + 3x^2 - 6x - 3x + 6 + 0.6667x^3 - 2.0001x^2 + 1.3334x - 2.0001x^2 + 4.0002x + 2.0001x - 4.0002$$

$$\rightarrow -0.01665x^5 - 0.04995x^4 - 0.00333x^3 + 0.0666x^4 - 0.1998x^3 + 0.1332x^2 - 0.08325x + 0.24973x^3 - 0.1665x^2 + 0.333x^3 - 0.999x^2 + 0.666x$$

$$f_5(x) = -0.01665x^5 - 0.7659x^4 + 0.34965x^3 - 1.0323x^2$$

(4-1)

$$f_5(x) = 0.01665x^5 - 0.25x^4 + 2.08333x^3 - 4.75x^2 + 5.9x - 5$$

~~✓~~

(4-2)

$$f'_5(x) = 0.08325x^4 - x^3 + 6.24999x^2 - 9.5x + 5.9$$

$$f'_5(1) = 1.43324$$

(4-3)

$$f''_5(x) = 0.333x^3 - 3x^2 + 12.49998x - 9.5$$

$$f''_5(3) = 9.99049$$

$$f''_5(3) = 9.99049$$

$$f'''_5(x) = 0.999x^2 - 6x + 12.49998$$

$$f'''_5(3) = 0.999(3)^2 - 6(3) + 12.49998$$

$$\Rightarrow f_3(x) = 0.999x^2 - 6x + 12.4998$$



[Handwritten signature]

(Q.5).

⇒ sol:-

$$\frac{dy}{dx} = 2x - y \quad / \quad y(0) = 1 \quad \begin{cases} y_0 = 1 \\ x_0 = 0 \end{cases}$$

$$x = 0.222 \quad / \quad h = 0.2$$

$$\Rightarrow k_1 = h f(x_i, y_i) = 0.2 f(0, 1) = 0.2 \times -1 = -0.2$$

$$\Rightarrow k_2 = h f(x_i + \frac{1}{2}h, y_i + \frac{1}{2}k_1) = 0.2 f(0.1, 0.9)$$

$$k_2 = 0.2 \times -0.7 = -0.14$$

$$\Rightarrow k_3 = h f(x_i + \frac{1}{2}h, y_i + \frac{1}{2}k_2) = 0.2 f(0.1, 0.93)$$

$$k_3 = 0.2 \times -0.73 = -0.146$$

$$\Rightarrow k_4 = h f(x_i + h, y_i + k_3) = 0.2 f(0.2, 0.854)$$

$$k_4 = 0.2 \times -0.454 = -0.0908$$

$\{y_i\}$

$$\Rightarrow y(0.2) = 1 \times \frac{1}{6} (-0.2 + (2 \times -0.14) + (2 \times -0.146) - 0.0908)$$

$$y(0.2) = 0.91987$$

Q.8 (m=2) ^{مساوي}

(8 values) ^{البيانات} poly (2nd) / r=0.9990 / st=2500
 $n=8$

$$S_{y/x} = \sqrt{\frac{S_r}{n-(m+1)}}$$

$$S_{y/x} = \sqrt{\frac{4.9975}{8-3}}$$

$$S_{y/x} = 0.99975$$

$$r = \sqrt{\frac{S_t - S_r}{S_t}}$$

$$0.9990 = \sqrt{\frac{2500 - S_r}{2500}}$$

$$S_r = 4.9975$$

$$S_y = \sqrt{\frac{S_t}{n-1}} = \sqrt{\frac{2500}{7}}$$

$$S_y = 18.89822$$

(standard deviation)

مستوى مكتوب
 على التوزيع
 بشيء
 (؟)

Am

Q.9.

$$f(x, y) = 2x^2 + 5y^2 + 4xy - 3x - 2y$$

(1, -1) , $\{ (0.5A) \leftarrow \text{check } \delta k \} \leftarrow \text{or}$

$\Rightarrow$ solve

$$\frac{\partial f}{\partial x} = 4x + 4y - 3 \quad / \quad \frac{\partial f}{\partial y} = 10y + 4x - 2$$

$$\nabla f = \begin{bmatrix} -3 \\ -8 \end{bmatrix} \quad / \quad \begin{aligned} x_{i+1} &= x_i + \delta_{ix} h = 1 - 3h \\ y_{i+1} &= y_i + \delta_{iy} h = -1 - 8h \end{aligned}$$

or δ_i

$$\Rightarrow g(h) = 2(1-3h)^2 + 5(-1-8h)^2 + 4(1-3h)(-1-8h) - 3(1-3h) - 2(-1-8h)$$

$$= 2(1 - 6h + 9h^2) + 5(1 + 16h + 64h^2) + 4(-1 + 8h + 3h + 24h^2) - 3 + 9h + 2 + 16h$$

$$= 2 - 12h + 18h^2 + 5 + 80h + 320h^2 - 4 + 32h + 96h^2 - 3 + 9h + 2 + 16h$$

$$g(h) = 434h^2 + 137h + 2 \quad / \quad g'(h) = 868h + 137 = 0$$

~~Ibrahim.kh~~ $\Rightarrow h = \frac{-137}{868} = -0.15783 / (1.47349 / 0.26264)$

SHOW YOUR DETAILED SOLUTIONS AND CALCULATIONS - Final answers

with no solution will get ZERO

Carry your intermediate computations to at least three decimal places.

**Q1- Use the golden-section search to find the minimum of $y = x^2 - 4\sin(5x)$
assume that $x_l = 1.3$ and $x_u = 1.7$ (10 marks)**

Answer:

.....

Q2- Solve the integral $\int_0^1 \frac{3}{1+x^2} dx$ by Trapezoidal rule with $n = 1, 2, 4$ and 8. Then obtain the better estimate using Romberg integration method. (13 marks)

Answer:

.....

**Q3- Consider the ordinary differential equation: $2 \frac{dy}{dx} - 3e^x + 2y = 0$
Given: $y(0) = 0, h = 0.5$. Use the Runge-Kutta method to the 4th order to find $y(1)$.
(12 marks)**

Answer:

.....

Q4: From the following data shown in the table below (15 marks)

$x:$	-1	0	1	3	5
$f(x)$	3.1	0	4.3	7.5	2.2

1- Find $f(2)$ - employing Newton's forward interpolation formula.

Answer:

.....

2- Find the area bounded by the curve and x axis from $x = -1$ to $x = 5$ using

Simpson's 1/3 rule

Answer:

.....

3- Find $f'(3)$ -using two points Centered finite-divided difference formula taking the data from the table.

Answer:

.....

End of questions

3

(minimum) (aleb)

الآلات راز
بلاش نشو و نما
از لایق

$$\Rightarrow d = 0.247 \text{ \AA}$$

$$\Rightarrow x_2 = x_w - d \Rightarrow x_w = 1.7 - 0.247 \Rightarrow x_2 = 1.453$$

$$\Rightarrow f(x_2) = (1.453)^2 - [4 \sin(5 \times 1.453)] \Rightarrow \boxed{f(x_2) = -1.215} \times$$

⇒ new interval $\rightarrow X_L = 1.453, X_U = 1.7$

[2 iterations]

$$\Rightarrow \boxed{d = 0.153} \times$$

$$\Rightarrow x_2 = x_u - \delta \Rightarrow x_2 = 1.7 - 0.153 \Rightarrow \boxed{x_2 = 1.547}$$

$$\Rightarrow f(x_2) = (1.547)^2 - [4 \sin(5 \times 1.547)] \Rightarrow \boxed{f(x_2) = -1.579}$$

min ✗

$$\Rightarrow [f(x_1) > f(x_2)] \Rightarrow x_u = x_1 \text{ \& } x_l = x_2 \text{ ✗}$$

من اليسار

$$\Rightarrow \text{New Interval} \rightarrow [x_u = 1.606, x_l = 1.453] \text{ ✗}$$

$$\Rightarrow \delta = 0.61803 (x_u - x_l) \Rightarrow \delta = 0.61803 (1.606 - 1.453)$$

$$\Rightarrow \boxed{\delta = 0.095} \text{ ✗}$$

$$\Rightarrow x_1 = x_l + \delta \Rightarrow x_1 = 1.453 + 0.095 \Rightarrow \boxed{x_1 = 1.548} \text{ ✗}$$

$$\Rightarrow x_2 = x_u - \delta \Rightarrow x_2 = 1.606 - 0.095 \Rightarrow \boxed{x_2 = 1.511} \text{ ✗}$$

$$\Rightarrow f(x_1) = (1.548)^2 - [4 \sin(5 \times 1.548)] \Rightarrow \boxed{f(x_1) = -1.578} \text{ ✗}$$

$$\Rightarrow f(x_2) = (1.511)^2 - [4 \sin(5 \times 1.511)] \Rightarrow \boxed{f(x_2) = -1.539} \text{ ✗}$$

min ✗

$$\Rightarrow [f(x_2) > f(x_1)] \Rightarrow x_l = x_2 \text{ \& } x_u = x_u \text{ ✗}$$

من اليمين

$$\Rightarrow \text{New Interval} \Rightarrow [x_l = 1.511, x_u = 1.606]$$

~~مع~~ ✗

~~breakdown~~

Q.23. $\left\{ \int_0^1 \frac{3}{1+x^2} \cdot dx \right\}$

⇒ sol:-

$f(x) = \frac{3}{1+x^2}$ / $n=1, 2, 4, 8 \Rightarrow 3^{\text{rd}} \text{ order.}$

$\{1 - (n-1) \text{ عدد} \leftarrow \text{تساوي}\}$

⇒ ~~$n=1$~~ $n=1.$

⇒ $h = \frac{b-a}{n} \Rightarrow h = \frac{1-0}{1} = \boxed{1}$

⇒ $R(1,0) = \frac{h}{2} [f(0) + f(1)]$

$= \frac{1}{2} [3 + 1.5]$

⇒ $R(1,0) = 2.25$

(i, k)	2 nd	2 nd	3 rd
$R(1,0)$	2.25	—	—
$R(2,0)$	$R(2,1)$	2.25	—
$R(3,0)$	$R(3,1)$	$R(3,2)$	—
$R(4,0)$	$R(4,1)$	$R(4,2)$	$R(4,3)$

لذلك في الجدول

😊

⇒ ~~$n=2$~~ $n=2.$

⇒ $h = \frac{b-a}{n} \Rightarrow h = \frac{1-0}{2} = \boxed{0.5}$

⇒ $R(2,0) = \frac{h}{2} [f(0) + f(1) + (2 \times f(0.5))]$

$= \frac{0.5}{2} [3 + 1.5 + (2 \times 2.4)]$

$R(2,0) = 2.325$

⇒ $n=4.$

⇒ $h = \frac{b-a}{n} \Rightarrow h = \frac{1-0}{4} = \boxed{0.25}$

⇒ $R(3,0) = \frac{h}{2} [f(0) + f(1) + 2(f(0.25) + f(0.5))]$

⇒ $R(3,0) = \frac{0.25}{2} [3 + 1.5 + 2(2.823 + 2.4)]$

⇒ $R(3,0) = 2.348$

$$\Rightarrow n = 8$$

$$\Rightarrow h = \frac{b-a}{n} \Rightarrow h = \frac{1-0}{8} = 0.125$$

$$\Rightarrow R(4,0) = \frac{h}{2} \left[f(0) + f(1) + 2 \left(f(0.125) + f(0.25) + f(0.375) + f(0.5) + f(0.625) + f(0.75) + f(0.875) \right) \right]$$

$$\Rightarrow R(4,0) = \frac{0.125}{2} \left[3 + 1.5 + 2(2.954 + 2.823 + 2.63 + 2.4 + 2.157 + 1.92 + 1.699) \right]$$

$$\Rightarrow R(4,0) = 2.354$$

قانون رونج-كوتس

$$\Rightarrow R(2,1) = \frac{4^k \cdot R(2,0) - R(1,0)}{4^1 - 1} \Rightarrow R(2,1) = \frac{(4 \cdot 2.325) - (1.75)}{3}$$

$$\Rightarrow R(2,1) = 2.35$$

$$\Rightarrow R(3,1) = \frac{4^k \cdot R(3,0) - R(2,0)}{4^1 - 1} \Rightarrow R(3,1) = \frac{(4 \cdot 2.348) - (2.325)}{3}$$

$$\Rightarrow R(3,1) = 2.356$$

$$R(4,1) = \frac{4^k \cdot R(4,0) - R(3,0)}{4^1 - 1} \Rightarrow R(4,1) = \frac{(4 \cdot 2.354) - 2.348}{3}$$

$$\Rightarrow R(4,1) = 2.356$$

$$\Rightarrow R(3,2) = \frac{(4^2 \times R(3,1)) - R(2,1)}{4^2 - 1}$$

$$\Rightarrow R(3,2) = \frac{(16 \times 2.356) - 2.35}{15} \Rightarrow \boxed{R(3,2) = 2.356} //$$

$$\Rightarrow R(4,2) = \frac{(4^2 \times R(4,1)) - R(3,1)}{15}$$

$$\Rightarrow R(4,2) = \frac{(16 \times 2.356) - (2.356)}{15} \Rightarrow \boxed{R(4,2) = 2.356} //$$

$$\Rightarrow R(4,3) = \frac{(4^3 \times R(4,2)) - R(3,2)}{63}$$

$$\Rightarrow R(4,3) = \frac{(64 \times 2.356) - 2.356}{63}$$

$$\Rightarrow R(4,3) = \boxed{2.356} //$$

Ibrahim Kh

Alc

[Q.3] .

⇒ 50/5

$$f(x_i, y_i)$$

$$\frac{2dy}{dx} - 3e^x + 2y = 0 \Rightarrow \left[\frac{dy}{dx} = \frac{3e^x - 2y}{2} \right]$$

$y(0) = 0 \xrightarrow{\text{Logic}} x_i = 0, y_i = 0 \quad / \quad h = 0.5 \quad / \quad y(1)$
 $y(0.5)$
 $y(1)$

(2 step) $\therefore \leftarrow$ (else) $\rightarrow$ {ok 4}

$$\Rightarrow k_1 = h^2(x_0, y_0) \Rightarrow k_1 = 0.5^2(0,0) \Rightarrow k = 0.5 \times 1.5 \Rightarrow k_1 = 0.75$$

$$\Rightarrow k_2 = h f\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right) \Rightarrow k_2 = 0.5 f(0.25, 0.375) \Rightarrow k_2 = 0.5 * 1.551$$

$$\Rightarrow K_2 = 0.775 *$$

$$\Rightarrow k_3 = hf(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}) \Rightarrow k_3 = 0.5 f(0.25, 0.387) \Rightarrow k_3 = 0.5 \cdot 4.539$$

$$\Rightarrow k_3 = 0.769 \text{ \#}$$

$$\Rightarrow k_x = h f(x_0 + h, y_0 + k_3) \Rightarrow k_x = 0.5 f(0.5, 0.769) \Rightarrow k_x = 0.5 \cdot 1.704$$

$$\Rightarrow K_{\text{eff}} = 0.852 \quad \#$$

$$\Rightarrow y(0.5) = y(0) + \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$

$$\Rightarrow y(0.5) = 0 + \frac{1}{6} (0.75 + (2 \times 0.775) + (2 \times 0.769) + 0.852)$$

$y(0.5) = 0.782$ * $\Rightarrow$ step (2) $\rightarrow$ done

$$\Rightarrow \text{step (2)} \rightarrow [x_0 = 0.5 / y_0 = 0.782] \neq$$

$$\Rightarrow k_1 = h f(x_0, y_0) \Rightarrow k_1 = 0.5 f(0.5, 0.782) \Rightarrow k_1 = 0.5 \times 1.691$$

$$\Rightarrow \boxed{k_1 = 0.845} \neq$$

$$\Rightarrow k_2 = h f\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right) \Rightarrow k_2 = 0.5 f(0.75, 1.204)$$

$$\Rightarrow k_2 = 0.5 \times 1.971 \Rightarrow \boxed{k_2 = 0.985} \neq$$

$$\Rightarrow k_3 = h f\left(x_0 + h, y_0 + k_2\right) \Rightarrow k_3 = 0.5 f(0.75, 1.274)$$

$$\Rightarrow k_3 = 0.5 \times 1.901 \Rightarrow \boxed{k_3 = 0.950} \neq$$

$$\Rightarrow k_4 = h f(x_0 + h, y_0 + k_3) \Rightarrow k_4 = 0.5 f(1, 1.732)$$

$$\Rightarrow k_4 = 0.5 \times 2.345 \Rightarrow \boxed{k_4 = 1.172} \neq$$

$$\Rightarrow y(1) = y(0.5) + \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$

$$\Rightarrow y(1) = 0.782 + \frac{1}{6} (0.845 + (2 \times 0.985) + (2 \times 0.950) + 1.172)$$

$$\Rightarrow \boxed{y(1) = 1.730} \neq \Rightarrow \text{step (2)} \rightarrow \underline{\underline{\text{done}}}$$

Ibrahim Kh...

Amr

(Q.4)

x	-1	0	1	3	5
f(x)	3.1	0	4.3	7.5	2.2

(1). $\Rightarrow$ sol:-

x_i	y_i	b_1	b_2	b_3	b_4
-1	3.1	-3.1	3.7	-1.15	0.186
0	0	4.3	-0.9	-0.032	
1	4.3	1.6	-1.062		
3	7.5	-2.65			
5	2.2				

ملاحظات للتذكير:-

III عند (3) نقال 3 ارقام اوقف

عند الدرجة (4) $\leftarrow$ وبالتالي

2 اوقف عند

(b₄)

$$\Rightarrow b_{1(1)} = \frac{3.1 - 0}{-1 - 0} = -3.1$$

$$\Rightarrow b_{1(2)} = \frac{0 - 4.3}{0 - 1} = 4.3$$

$$\Rightarrow b_{1(3)} = \frac{4.3 - 7.5}{1 - 3} = 1.6$$

$$\Rightarrow b_{1(4)} = \frac{7.5 - 2.2}{3 - 5} = -2.65$$

$$\Rightarrow b_{2(1)} = \frac{-3.1 - 4.3}{-1 - 1} = 3.7$$

$$\Rightarrow b_{2(2)} = \frac{4.3 - 1.6}{0 - 3} = -0.9$$

$$\Rightarrow b_{2(3)} = \frac{1.6 - 2.65}{1 - 5} = -1.062$$

$$\Rightarrow b_{3(1)} = \frac{3.7 + 0.9}{-1 - 3} = -1.15$$

$$\Rightarrow b_{3(2)} = \frac{-0.9 + 1.062}{0 - 5} = -0.032$$

$$b_4 = \frac{-1.15 + 0.032}{-1 - 5}$$

$$b_4 = 0.186$$

$$\Rightarrow f(x) = b_0 + b_1(x-x_0) + b_2(x-x_0)(x-x_1) + b_3(x-x_0)(x-x_1)(x-x_2) \\ + b_4(x-x_0)(x-x_1)(x-x_2)(x-x_3)$$

دالة رانك بول

$$\Rightarrow f(2) = 3.1 + [-3.1(2+1)] + [3.7(2+1)(2-0)] \\ + [-1.15(2+1)(2-0)(2-1)] + [0.186(2+1)(2-0)(2-1)(2-3)]$$

$$\Rightarrow \boxed{f(2) = 7.984}$$

Ans

Ibrahim

(3). $\Rightarrow$ solve

$f'(3) \xrightarrow{\text{use}}$ center & first dir. — (مشتق أولي)

$$\Rightarrow f'(3) = \frac{f(x_{i+1}) - f(x_{i-1}))}{2h}$$

(٢) كيف بدو أعرفها $\leftarrow$ مدجنا هو الجدول لأننا هو الأساس (السؤال ومنه) ال (٣) وقد عرفنا بنحوها.

$\Rightarrow$ هو بدو $\leftarrow f'(3) \leftarrow$ كيف $\leftarrow$ ال (3) بين عين وميلين و/أو قاعلا

1	3	5
4.3	7.5	2.2

$$\Rightarrow h = 2$$

$$\Rightarrow f'(3) = \frac{f(5) - f(1)}{2 \times 2} = \frac{2.2 - 4.3}{4}$$

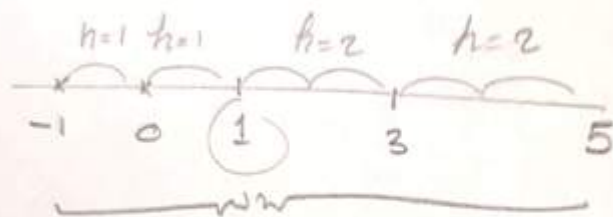
$$\Rightarrow f'(3) = -0.525$$

~~Ibrahim~~

دعواتكم للأهلنا
في غزوة الحرة



$$\underline{n=4}$$



$$\underline{n=2}$$

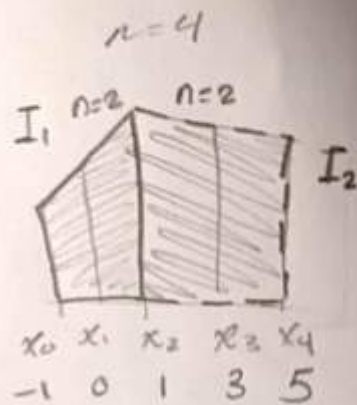
$$I_1 = \frac{h}{3} [f(-1) + f(1) + 4f(0)]$$

$$h=1 \quad \frac{1}{3} (3.1 + 4.3 + 0)$$

$$= 2.4666$$

$$I_2 = \frac{h}{3} [f(1) + f(5) + 4f(3)]$$

$$h=2 \quad I_2 \quad \frac{2}{3} * (4.3 + 2.2 + 4(7.5)) = 24.333$$



$$\Sigma I = 26.799$$

$$f'_{eq} = f'_{eq,1} + f'_{eq,2}$$

Exam Time: 90 minutes

SHOW YOUR DETAILED SOLUTIONS AND CALCULATIONS. Final answer with a short note at the end.

Answer the following questions

Q1- Fit a straight-line to the data provided in the following Table

(10 marks)

x_i	5	7	9	11	13
y_i	6	4.5	3.4	2.3	1.4

The equation of the line is:

The value of coefficient of determination, r^2 is:

Q2- Compute the integration: $\int_2^4 \frac{1}{(x-1)^2} dx$, using Work to five significant figures

(10 marks)

Apply Romberg integration taking $n=1$, $n=2$ & $n=4$ up to $R(3, 2)$.

Answer:

Q3- Evaluate the following double integration: $\int_1^3 \int_1^3 (2x + y) dx dy$.

(10 marks)

Take $dx = 0.5$ in the x -direction, $dy = 0.5$ in the y -direction. Apply the Trapezoidal rule in the x -direction and in the y -direction.

Answer:

Q4- Maximize the function: $f(x, y) = xy + 4x - x^2 - y^2 - 4$

(10 marks)

Apply the Optimal steepest ascent method, use point (2, 2) as the initial estimate of the optimal solution. The optimum point after conducting one iteration is:

Answer:

Q5- Consider the ordinary differential equation: $\frac{dy}{dx} = x^2 - 2y^2 + x$

(10 marks)

Given: $x_0 = 1$, $y_0 = 2$, $h = 0.5$

Use the Runge-Kutta method to the 4th order to find $y(1.5)$.

Answer:

End of Questions

Q(1) straight-line The equation of the line is: $y = 8.51 - 0.55x$

x_i y_i $R^2 = 0.9844 \Rightarrow 98.44\%$

• 5 6 sol $n = 5$

• 7 4.5 $\sum x^2 = 445$ $\sum x = 45$ $\bar{x} = \frac{\sum x}{n} = \frac{45}{5} = 9$
 • 9 3.4 $\sum y^2 = 75.66$ $\sum y = 17.8$ $\bar{y} = \frac{\sum y}{n} = \frac{17.8}{5} = 3.56$
 • 11 2.3 $\sum xy = 138.2$ $\sum x^3 = 4725$
 • 13 1.6 $\sum x^2 y = 1194.6$ $\sum x^4 = 52789$

$y = A + Bx$ $A = 8.51$ $B = -0.55$ $r = 0.99216$

$\boxed{a_1} = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{n \sum x_i^2 - (\sum x_i)^2} = \frac{(5)(138.2) - (45)(17.8)}{(5)(445) - (45)^2} = -0.55$

$\boxed{a_0} = \bar{y} - a_1 \bar{x} = 3.56 - (-0.55)(9) = 8.51$

الحل
مكرر

Q(2) $\int_2^4 \frac{1}{(x-1)^2} dx$ Apply Romberg integration

$n=1, n=2$ & $n=4$ up to (3,2)

<u>sol</u> Zero or	1 st order	2 nd order
$R(1,0)$		
$R(2,0)$	$R(2,1)$	
$R(3,0)$	$R(3,1)$	$R(3,2)$

* $R(1,0)$: $h = \frac{b-a}{n} = \frac{4-2}{1} = 2$
2, 4

$f(2) = 1$ $I = \frac{h}{2} (f(2) + f(4))$

$f(4) = \frac{1}{9}$ $= \frac{2}{2} (1 + \frac{1}{9}) = 1.111$

* $R(2,0)$: $h = \frac{b-a}{n} = \frac{2}{2} = 1$

2, 3, 4

$f(2) = 1$, $f(3) = 0.25$, $f(4) = \frac{1}{9}$

$I = \frac{1}{2} (1 + \frac{1}{9} + 2(0.25)) = 0.8056$

* $R(3,0)$: $h = \frac{b-a}{n} = \frac{2}{4} = \frac{1}{2} = 0.5$

2, 2.5, 3, 3.5, 4

$f(2) = 1$, $f(2.5) = \frac{4}{9}$, $f(3) = 0.25$, $f(3.5) = 0.16$, $f(4) = \frac{1}{9}$

$I = \frac{0.5}{2} (1 + \frac{4}{9} + 2(\frac{4}{9} + 0.25 + 0.16))$

$= 0.705$

$R(2,1) = \frac{4^1 R(2,0) - R(1,0)}{4^1 - 1}$

$= 0.7037$

$R(3,1) = \frac{4^1 R(3,0) - R(2,0)}{4^1 - 1}$

$= 0.671481$

$R(3,2) = \frac{4^2 R(3,1) - R(2,1)}{4^2 - 1}$

$= 0.66933$

الحل
مكرر

$$Q(3) \int_1^3 \int_1^3 (2x+y) dx dy$$

$$dx = 0.5$$

$$dy = 0.5$$

Apply Trapezoidal Rule x-direction and in the y-direction.

Sol

$y \backslash x$	1	1.5	2	2.5	3
1	3	4	5	6	7
1.5	3.5	4.5	5.5	6.5	7.5
2	4	5	6	7	8
2.5	4.5	5.5	6.5	7.5	8.5
3	5	6	7	8	9

$$x\text{-axis} \rightarrow h_1 = dx = 0.5$$

$$f_1 = \frac{0.5}{2} (3+7+2(4+5+6)) = \underline{10}$$

$$f_2 = \frac{0.5}{2} (3.5+7.5+2(4.5+5.5+6.5)) = \underline{11}$$

$$f_3 = \frac{0.5}{2} (4+8+2(5+6+7)) = \underline{12}$$

$$f_4 = \frac{0.5}{2} (4.5+8.5+2(5.5+6.5+7.5)) = \underline{13}$$

$$f_5 = \frac{0.5}{2} (5+9+2(6+7+8)) = \underline{14}$$

↓ y-axis

$$I = \frac{0.5}{2} (10+14+2(11+12+13)) = \underline{24}$$

الحل هو 24

$$Q(4) f(x,y) = xy + 4x - x^2 - y^2 - 4$$

Apply the optimal steepest ascent method.

Sol $(2, 2)$
 x_i y_i

$$(2, 2)$$

one iteration.

$$\frac{df}{dx} = y + 4 - 2x = 2$$

$$\nabla f(2,2) = \begin{bmatrix} 2 \\ -2 \end{bmatrix}$$

$$\frac{df}{dy} = x - 2y = -2$$

$$x_{i+1} = 2 + 2h$$

$$y_{i+1} = 2 - 2h$$

$$f(2+2h, 2-2h) = (2+2h)(2-2h) + 4(2+2h) - (2+2h)^2 - (2-2h)^2 - 4$$

$$4 - 4h + 4h - 4h^2 + 8 + 8h - (4 + 8h + 4h^2) - (4 - 8h + 4h^2) - 4$$

$$\underline{4} - \underline{4h} + \underline{4h} - \underline{4h^2} + \underline{8} + \underline{8h} - \underline{4} - \underline{8h} - \underline{4h^2} - \underline{4} + \underline{8h} - \underline{4h^2} - \underline{4}$$

$$-12h^2 + 8h = g(x)$$

$$-24h + 8 = 0$$

$$g'(x) = -24h + 8$$

$$\frac{24h}{24} = \frac{8}{24}$$

$$h = \frac{1}{3}$$

$$0.333$$

$$\left(\frac{8}{3}, \frac{4}{3} \right)$$

$$(2.666, 1.333)$$

الحل هو 2.666, 1.333

Q(5) O.D.E $\frac{dy}{dx} = x^2 - 2y^2 + x$ $x_0 = 1, y_0 = 2, h = 0.5$ $y(1) = 2$

Use the R.K method to the 4th order to find $y(1.5)$

$$h = 1.5 - 1 = \underline{\underline{0.5}}$$

sol

$$K_1 = 0.5 f(1, 2) = 0.5 \times -6 = \underline{\underline{-3}}$$

$$K_2 = 0.5 f\left(1 + \frac{0.5}{2}, 2 + \frac{-3}{2}\right) = 0.5 f(1.25, 0.5) = 0.5 \times 2.3125 = \underline{\underline{1.15625}}$$

$$K_3 = 0.5 f\left(1.25, 2 + \frac{1.15625}{2}\right) = 0.5 f(1.25, 2.578125) = 0.5 \times -10.481 = \underline{\underline{-5.2405}}$$

$$K_4 = 0.5 f(1.5, -3.2405) = 0.5 \times -17.2517 = \underline{\underline{-8.62585}}$$

$$y(1.5) = 2 + \frac{1}{6} (-3 + 2(1.15625) + 2(-5.2405) + (-8.62585))$$

$$\boxed{y(1.5) = -1.299058} \quad \times$$

الحلوة